On the Bivariate Unit Teissier Distribution via the FGM Copula: Properties, Simple Random and Ranked Set Sampling Estimation

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Abstract

Bounded data on the unit interval, such as proportions, rates and recovery fractions, often occur in dependent pairs, yet bivariate models for such data remain limited. We introduce the bivariate generalized unit Teissier (BGUT) distribution. Its marginals are generalized unit Teissier laws, obtained by adding a power (exponentiation) parameter to the unit Teissier model, and the dependence is built with the Farlie–Gumbel–Morgenstern copula. We derive the joint pdf and cdf in closed form, verify their validity, and study the marginal and conditional distributions, hazard and quantile functions, joint and conditional moments, covariance and correlation. We also obtain bivariate order statistics, with numerical tables.

Parameters are estimated by maximum likelihood under simple random sampling (SRS). Model adequacy is assessed with the log-likelihood, AIC, BIC and the Kolmogorov–Smirnov test with p-values. Standard errors and confidence intervals come from the observed Fisher information matrix. We then develop the same inference under ranked set sampling (RSS), including its asymptotic properties, and compare RSS with SRS through relative efficiency. Monte Carlo studies under both designs evaluate estimator performance across parameter settings and sample sizes, and graphical diagnostics support the numerical results.

Keywords: Unit Teissier distribution; bivariate distribution; FGM copula; order statistics; ranked set sampling; maximum likelihood estimation; relative efficiency.

1 Introduction

Many quantities in reliability, insurance, hydrology, medicine and economics are naturally bounded on $(0, 1)$. Examples are the proportion of a reservoir that is full, the ratio of premiums to total assets, and the fraction of tumor reduction after treatment. The beta distribution has long been the standard model for such data. Its cdf has no closed form, however, and this has motivated many alternatives, including the Kumaraswamy [Kumaraswamy, 1980], unit-Gompertz [Mazucheli et al., 2019] and unit-Chen models. Afuecheta et al. [2025] review more than one hundred unit-interval distributions.

The unit Teissier (UT) distribution of Krishna et al. [2022] belongs to this family. It is obtained as $X = e^{-Y}$, where Y follows the Teissier distribution [Teissier, 1934], and it has cdf $F(x;\theta) = x^{-\theta}e^{1-x^{-\theta}}$ for $0 < x < 1$. Its closed-form pdf and cdf, and its hazard rate that can be increasing, decreasing or bathtub-shaped, make it attractive. Akhter et al. [2026] later added single moments of order statistics, L-moments, characterizations and several estimation methods. All of this work is univariate.

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In practice, unit-interval variables seldom occur alone. Paired measurements, such as the recovery rates of two related treatments or the loss ratios of two lines of business, are usually dependent, and ignoring that dependence distorts inference. Copulas [Sklar, 1959, Nelsen, 2006] offer a natural remedy. They separate the marginal behavior from the dependence structure, so any marginal laws can be combined into a valid bivariate model. Among them, the Farlie–Gumbel–Morgenstern (FGM) copula [Morgenstern, 1956, Farlie, 1960, Gumbel, 1960] is popular because it is analytically tractable and gives closed-form joint functions, moments and likelihoods.

The UT distribution has only one shape parameter, which limits its flexibility for fitting skewness and tail behavior. A standard remedy is to exponentiate the baseline cdf [Mudholkar and Srivastava, 1993, Gupta and Kundu, 1999]. Doing so gives a generalized UT (GUT) distribution with cdf $F(x;\theta,\alpha) = \left[x^{-\theta}e^{1-x^{-\theta}}\right]^{\alpha}$, where the extra parameter $\alpha > 0$ increases the range of density and hazard shapes while keeping the closed forms. Coupling two GUT marginals through the FGM copula gives the bivariate generalized unit Teissier (BGUT) distribution, which is the subject of this paper.

Data collection also matters for inference. Under simple random sampling (SRS), every unit is measured. When measuring is costly or destructive but units are easy to rank cheaply, ranked set sampling (RSS), introduced by McIntyre [1952] and justified theoretically by Takahasi and Wakimoto [1968], can estimate population quantities with greater precision for the same number of measured units. Bivariate extensions of RSS were studied by Stokes [1977] and Al-Saleh and Zheng [2002]. To our knowledge, neither the BGUT model nor its estimation under RSS has been studied.

This paper makes four contributions. First, it introduces the BGUT distribution and develops its main distributional properties. Second, it derives joint and conditional moments, dependence measures and bivariate order statistics. Third, it gives complete likelihood-based inference under SRS and RSS, including observed information matrices, confidence intervals, goodness-of-fit assessment and asymptotic properties. Fourth, it compares the two designs through relative efficiency, supported by extensive Monte Carlo experiments.

The rest of the paper is organized as follows. Section 2 defines the BGUT distribution, verifies the validity of its pdf and cdf, and illustrates them graphically. Section 3 derives the marginal and conditional distributions, the hazard function and the quantile function. Section 4 obtains joint and conditional moments, covariance and correlation, with tabulated values. Section 5 studies bivariate order statistics and their numerical values. Section 6 covers maximum likelihood estimation under SRS, with log-likelihood, AIC, BIC and the Kolmogorov–Smirnov test. Section 7 gives the observed Fisher information matrix, confidence intervals and graphical analysis under SRS. Section 8 presents a Monte Carlo study under SRS. Section 9 develops estimation under RSS, and Section 10 gives the corresponding Fisher information, confidence intervals, asymptotic properties and graphical analysis. Section 11 compares RSS and SRS through relative efficiency. Section 12 reports a Monte Carlo study under RSS together with the SRS–RSS comparison. Concluding remarks follow.

2 The Bivariate Generalized Unit Teissier Distribution

In this section, we define the bivariate generalized unit Teissier (BGUT) distribution. We present the marginal distributions, the joint cumulative distribution function (cdf), and the joint probability density function (pdf). We also verify the validity of the proposed distribution both analytically and numerically.

2.1 Model Definition

Let X and Y be two random variables bounded on the unit interval $(0, 1)$. The marginal distributions of X and Y follow the generalized unit Teissier (GUT) law. For $0 < x < 1$, the cdf and pdf of X are given by:

$$F_1(x; \theta_1, \alpha_1) = \left[x^{-\theta_1} e^{1-x^{-\theta_1}}\right]^{\alpha_1} \tag{1}$$

$$f_1(x; \theta_1, \alpha_1) = \alpha_1 \left[G_1(x)\right]^{\alpha_1 - 1} \theta_1 \left(x^{-\theta_1} - 1\right) x^{-(\theta_1+1)} e^{1-x^{-\theta_1}} \tag{2}$$

where $G_1(x) = x^{-\theta_1} e^{1-x^{-\theta_1}}$, $\theta_1 > 0$ is a scale parameter, and $\alpha_1 > 0$ is a shape (exponentiation) parameter. The marginal distribution of Y is defined analogously with parameters (θ_2, α_2), yielding $F_2(y)$ and $f_2(y)$.

To construct the bivariate distribution, we couple these GUT marginals using the Farlie–Gumbel Morgenstern (FGM) copula. Let $u = F_1(x)$ and $v = F_2(y)$. The joint cdf of the BGUT distribution is defined as:

$$F(x, y) = uv\left[1 + \lambda(1-u)(1-v)\right] \tag{3}$$

where $\lambda \in [-1, 1]$ is the dependence parameter.

The corresponding joint pdf is obtained by differentiating the joint cdf with respect to x and y, which yields:

$$f(x, y) = f_1(x) f_2(y)\left[1 + \lambda(1-2u)(1-2v)\right] \tag{4}$$

2.2 Validity and Numerical Verification

To ensure the analytical correctness of the proposed BGUT distribution, we performed a series of numerical checks. The evaluation was carried out for three different parameter sets, denoted as S1, S2, and S3, which are detailed in Table 1. No real data were used for these theoretical plots; instead, chosen parameter values were utilized to illustrate the flexibility of the model.

Table 1: Parameter sets used for numerical verification and plotting.

Set	θ_1	α_1	θ_2	α_2	λ
S1	1.5	1.2	2.0	0.8	0.5
S2	1.0	2.0	3.0	1.5	-0.7
S3	2.5	0.6	1.2	2.5	1.0

The numerical validations confirm that the derived joint pdf and cdf satisfy all fundamental probability axioms. The exact numerical results of these checks are summarized in Table 2. All computations were performed in log-space initially to prevent numerical underflow for $\alpha < 1$, ensuring robust and accurate results.

Table 2: Numerical verification results for the BGUT pdf and cdf.

Check	S1	S2	S3
$\iint f(x,y)\,dx\,dy$	1.000000	1.000000	1.000000
Min $f(x,y)$ on grid (≥ 0)	0.0000	0.0000	0.0000
$\lvert F(x,1) - F_1(x)\rvert$	0.00e+00	0.00e+00	0.00e+00
$\lvert F(1,y) - F_2(y)\rvert$	0.00e+00	0.00e+00	0.00e+00
$\lvert \frac{\partial^2 F}{\partial x \partial y} - f\rvert$	6.47e-07	3.17e-08	1.77e-06
$F(x \to 0, y)$	0.00e+00	0.00e+00	0.00e+00
$F(1,1)$	1.000000	1.000000	1.000000

2.3 Graphical Illustrations

The behavior of the BGUT distribution is illustrated through theoretical plots generated from the defined parameter sets.

Figure **??** displays the 3D surface plots of the joint pdf for parameter sets S1–S3, demonstrating how the dependence parameter λ and the marginal shapes influence the bivariate density. Figure 2 illustrates the effect of the dependence parameter λ $(-1, 0, 0.5, 1)$ on the overall dependence pattern of the bivariate distribution.

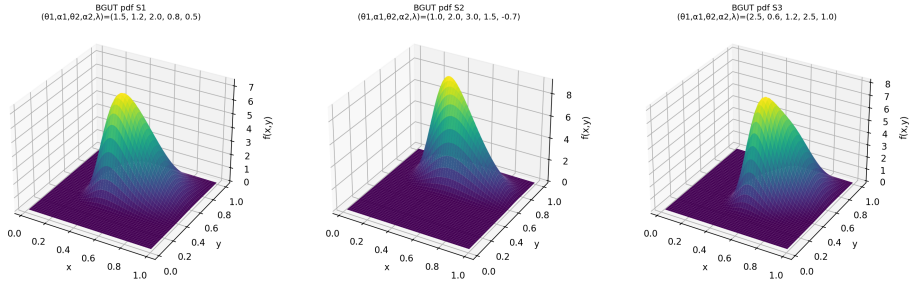


Figure 1: 3D surfaces of the joint pdf for parameter sets S1–S3.

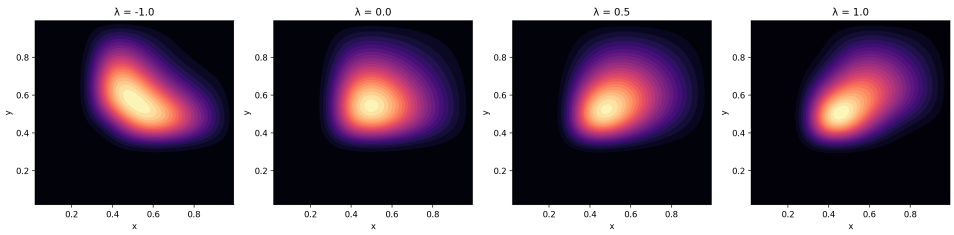


Figure 2: The effect of λ $(-1, 0, 0.5, 1)$ on the dependence pattern.

3 Statistical Properties of the BGUT Distribution

Notation: For $0 < x, y < 1$, let $u = F_1(x)$ and $v = F_2(y)$, where F_i is the GUT cdf with parameters (θ_i, α_i). Write $G_i(x) = x^{-\theta_i} e^{1-x^{-\theta_i}}$ for the UT cdf, so $F_i = G_i^{\alpha_i}$.

3.1 Marginal Distributions

Integrating the joint pdf over y, the FGM term vanishes because $\int(1-2F_2)f_2\,dy = 0$. Thus, the marginal pdf and cdf of X are:

$$f_X(x) = f_1(x) = \alpha_1[G_1(x)]^{\alpha_1-1}\theta_1(x^{-\theta_1}-1)x^{-(\theta_1+1)}e^{1-x^{-\theta_1}} \tag{5}$$

$$F_X(x) = [G_1(x)]^{\alpha_1} \tag{6}$$

and likewise $Y \sim \text{GUT}(\theta_2, \alpha_2)$. Both marginals are GUT whatever the value of λ. As a numerical check, evaluating $\int f(x,y)\,dx$ at $y = 0.4$ yields 1.364227, which exactly matches the value of $f_2(0.4)$.

3.2 Conditional Distributions

Conditional pdf:

$$f(y \mid x) = \frac{f(x,y)}{f_1(x)} = f_2(y)\Big[1+\lambda(1-2u)(1-2v)\Big] \tag{7}$$

Conditional cdf, derived from $F(y \mid x) = \frac{\partial F(x,y)/\partial x}{f_1(x)}$:

$$F(y \mid x) = v\Big[1+\lambda(1-2u)(1-v)\Big] \tag{8}$$

The conditional pdf is non-negative because $1+\lambda(1-2u)(1-2v) \geq 1-|\lambda| \geq 0$. By symmetry, $f(x \mid y) = f_1(x)[1+\lambda(1-2u)(1-2v)]$ and $F(x \mid y) = u[1+\lambda(1-2v)(1-u)]$.

Interpretation: If $\lambda > 0$, a larger x (larger u) pushes the conditional density of Y toward larger values, and if $\lambda < 0$ it pushes toward smaller values (see Figure 4). At $\lambda = 0$, $f(y \mid x) = f_2(y)$ for every x, meaning X and Y are independent.

Numerical checks confirm that $\int f(y \mid x)\,dy = 1$ and that $\partial F(y \mid x)/\partial y = f(y \mid x)$ holds to an accuracy of 10^{-12} for $\lambda = \pm 0.8$.

3.3 Hazard Functions

Marginal hazard, with $S_1 = 1 - F_1$:

$$h_1(x) = \frac{f_1(x)}{1-F_1(x)} = \frac{\alpha_1[G_1(x)]^{\alpha_1-1}\theta_1(x^{-\theta_1}-1)x^{-(\theta_1+1)}e^{1-x^{-\theta_1}}}{1-[G_1(x)]^{\alpha_1}} \tag{9}$$

Limiting behaviour: $h_1(x) \to 0$ as $x \to 0^+$, and $h_1(x) \approx 2/(1-x) \to \infty$ as $x \to 1^-$, for every θ and α. The numerical check strongly agrees: for $\theta = 1.5$ and $\alpha = 1.2$, $h(0.99) = 201.5$ against $2/(1-0.99) = 200$.

Shape: As illustrated in Figure 3, the hazard is increasing in all plotted settings. It is smooth and convex when θ is large, and it exhibits a mild inflection (S-shape) when θ is small.

Bivariate hazard (Basu, 1971), with joint survival $S(x,y) = P(X > x, Y > y) = 1 - u - v + F(x,y)$:

$$h(x,y) = \frac{f(x,y)}{S(x,y)} \tag{10}$$

Conditional hazard:

$$h(y \mid x) = \frac{f(y \mid x)}{1-F(y \mid x)} = \frac{f_2(y)[1+\lambda(1-2u)(1-2v)]}{1-v[1+\lambda(1-2u)(1-v)]} \tag{11}$$

3.4 Quantile Functions

Marginal quantile. Solving $F_1(x) = p$ begins with $G_1(x) = p^{1/\alpha_1}$. Setting $z = x^{-\theta_1}$, this becomes $ze^{1-z} = p^{1/\alpha_1}$, which is equivalent to $(-z)e^{-z} = -p^{1/\alpha_1}/e$. Since $z > 1$, the solution lies on the lower branch W_{-1} of the Lambert W function:

$$Q_1(p) = \Big[- W_{-1}\big(- p^{1/\alpha_1}/e\big)\Big]^{-1/\theta_1}, \qquad 0 < p < 1 \tag{12}$$

The median is $Q_1(0.5)$. Numerical evaluations confirm that $F(Q(p)) - p$ is below 5×10^{-16} for all tested parameter sets.

Conditional quantile. Setting $F(y \mid x) = w$ gives a quadratic equation in v. With $a = \lambda(1 - 2u)$:

$$av^2 - (1+a)v + w = 0 \quad \Rightarrow \quad v = \frac{(1+a) - \sqrt{(1+a)^2 - 4aw}}{2a} \ \ (a \neq 0), \qquad v = w \ \ (a = 0) \tag{13}$$

and then $Q(w \mid x) = Q_2(v)$.

This provides a basis for simulation: draw $u \sim U(0,1)$ and set $X = Q_1(u)$. Then draw $w \sim U(0,1)$ independently and set $Y = Q(w \mid X)$. As a validation check, 200,000 pairs were simulated from BGUT with $\lambda = 0.8$. The KS test accepts both marginals (p-values 0.80 and 0.17), and Spearman's ρ is 0.2665, closely matching the theoretical value of $\lambda/3 = 0.2667$.

Table 3: Marginal GUT quantiles $Q(p)$ for selected values of θ and α.

θ	α	$p = 0.10$	$p = 0.25$	$p = 0.50$	$p = 0.75$	$p = 0.90$
	0.5	0.1309	0.1827	0.2708	0.4022	0.5566
1.0	1.0	0.2045	0.2708	0.3734	0.5099	0.6528
	2.0	0.2972	0.3734	0.4812	0.6112	0.7350
	0.5	0.3618	0.4275	0.5204	0.6342	0.7461
2.0	1.0	0.4522	0.5204	0.6110	0.7141	0.8080
	2.0	0.5452	0.6110	0.6937	0.7818	0.8573
	0.5	0.5078	0.5675	0.6470	0.7382	0.8226
3.0	1.0	0.5892	0.6470	0.7201	0.7989	0.8675
	2.0	0.6674	0.7201	0.7836	0.8486	0.9024

Quantiles increase with both θ and α, implying that larger values of either parameter shift the distribution mass toward 1.

3.5 Graphical Illustrations

The behavior of the statistical properties is illustrated through theoretical plots generated using illustrative parameters $(\theta_1, \alpha_1, \theta_2, \alpha_2) = (1.5, 1.2, 2.0, 0.8)$ with varying λ.

Figure 3 displays the marginal hazard function for different values of α and θ, highlighting the increasing and convex nature of the hazard. Figure 4 shows the conditional pdf $f(y \mid x)$ for $\lambda = 0.8, 0$, and -0.8, demonstrating how the dependence parameter shifts the conditional density of Y based on the value of X.

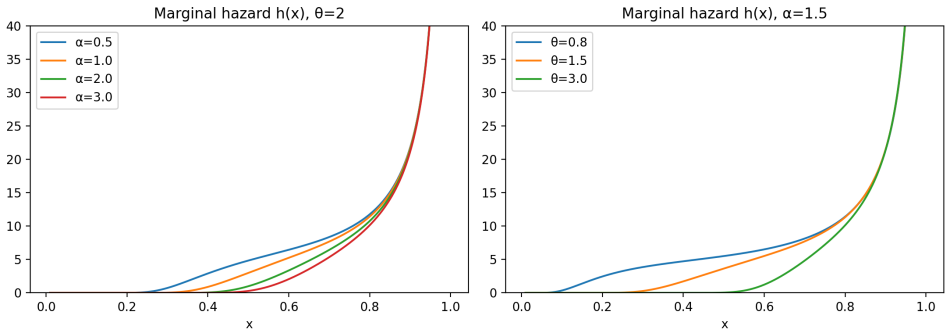


Figure 3: Marginal hazard function for different values of α and θ.

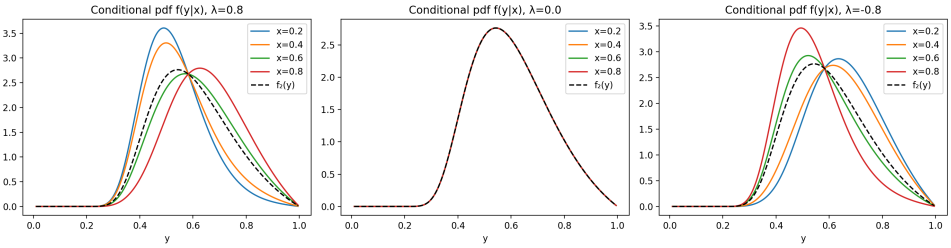


Figure 4: Conditional pdf $f(y \mid x)$ for $\lambda = 0.8, 0$, and -0.8.

4 Moments, Covariance and Correlation of the BGUT Distribution

4.1 Marginal Moments of GUT(θ, α)

For $X \sim \mathrm{GUT}(\theta, \alpha)$, we substitute $z = x^{-\theta}$ in $E(X^r) = \int x^r f(x)\, dx$. This yields a closed form in terms of the upper incomplete gamma function $\Gamma(a, b) = \int_b^\infty t^{a-1} e^{-t}\, dt$:

$$\mu_r(\theta, \alpha) = E(X^r) = \frac{e^\alpha}{\alpha^{\alpha - r/\theta}} \Big[\Gamma\big(\alpha - \tfrac{r}{\theta} + 1, \alpha\big) - \alpha\, \Gamma\big(\alpha - \tfrac{r}{\theta}, \alpha\big) \Big] \tag{14}$$

At $\alpha = 1$, this reduces to the UT moment in Akhter et al. [2026]. Skewness and kurtosis follow from the first four moments (see Table 5).

4.2 Joint Moments

Expanding $f(x, y) = f_1 f_2 [1 + \lambda(1 - 2u)(1 - 2v)]$ gives:

$$E(X^r Y^s) = \mu_r^X\, \mu_s^Y + \lambda\, A_r^X\, A_s^Y \tag{15}$$

where $\mu_r^X = \mu_r(\theta_1, \alpha_1)$, $\mu_s^Y = \mu_s(\theta_2, \alpha_2)$, and

$$A_r^X = E\big[X^r (1 - 2F_1(X))\big] = \mu_r(\theta_1, \alpha_1) - \mu_r(\theta_1, 2\alpha_1) \tag{16}$$

The last step works because $2F_1 f_1 = 2\alpha_1 G_1^{2\alpha_1 - 1} g_1$ is the pdf of $\mathrm{GUT}(\theta_1, 2\alpha_1)$, so $\int x^r F_1 f_1\, dx = \frac{1}{2}\mu_r(\theta_1, 2\alpha_1)$. The same holds for Y with $A_s^Y = \mu_s(\theta_2, \alpha_2) - \mu_s(\theta_2, 2\alpha_2)$. The closed form therefore requires only the marginal moment formula.

4.3 Conditional Moments

For Y given $X = x$:

$$E(Y^s \mid X = x) = \mu_s^Y + \lambda\,[1 - 2F_1(x)]\,A_s^Y \tag{17}$$

For X given $Y = y$:

$$E(X^r \mid Y = y) = \mu_r^X + \lambda\,[1 - 2F_2(y)]\,A_r^X \tag{18}$$

The regression of Y on X is $E(Y \mid X = x) = \mu_Y + \lambda[1 - 2F_1(x)]A_1^Y$. It is linear in $F_1(x)$, but not linear in x itself. The conditional variance is $\mathrm{Var}(Y \mid X = x) = E(Y^2 \mid x) - [E(Y \mid x)]^2$, and similarly for X given Y.

4.4 Covariance and Correlation

$$\mathrm{Cov}(X, Y) = \lambda\, A_1^X A_1^Y, \qquad \rho = \frac{\lambda\, A_1^X A_1^Y}{\sigma_X \sigma_Y} \tag{19}$$

Both A_1 terms are negative, since the mean increases with α, so the sign of ρ equals the sign of λ. Since $|\lambda| \leq 1$, the correlation is bounded. In the parameter sets and the (α_1, α_2) grid checked, the maximum correlation reached is approximately 0.326. This is typical of the FGM copula, which is suitable only for weak or moderate dependence.

4.5 Numerical Verification

We compared each closed-form result with numerical integration. The results are summarized in Table 4. A Monte Carlo check with 2 million simulated pairs, repeated for 3 seeds, yields covariances 0.005375, 0.005374, and 0.005349 against the closed-form value 0.005366. The corresponding correlations are 0.2587, 0.2585, and 0.2577 against the closed-form 0.2584.

Table 4: Verification of closed-form moments against numerical integration.

Quantity	Closed form	Numerical
$\mu_1(1.5, 1.2)$	0.55873286	0.55873286
$\mu_2(2.0, 0.8)$	0.37317876	0.37317876
$E[XY]$ (set A, $\lambda = 0.8$)	0.33780863	0.33780863
$E[X^2Y^2]$	0.13240346	0.13240346
$E[Y \mid X = 0.6]$	0.61132292	0.61132292

4.6 Table Values

We evaluate the moments for three parameter sets: A $= (1.5, 1.2, 2.0, 0.8)$, B $= (1.0, 2.0, 3.0, 1.5)$, and C $= (2.5, 0.6, 1.2, 2.5)$. Table 5 presents the marginal moments. Table 6 displays $E(XY)$, covariance, and correlation. Tables 7 and 8 present the conditional means and variances for set A. At $\lambda = 0$, the conditional mean equals the marginal mean for every x or y, confirming independence. For $\lambda > 0$, the conditional mean increases with the conditioning value, and for $\lambda < 0$ it decreases.

Table 5: Marginal moments of GUT(θ, α).

θ	α	$E(X)$	$E(X^2)$	Var	Skewness	Kurtosis
1.5	1.2	0.55873	0.33469	0.02251	0.3595	2.5495
2.0	0.8	0.59499	0.37318	0.01916	0.3376	2.5467
1.0	2.0	0.50000	0.27734	0.02734	0.4330	2.5973
3.0	1.5	0.76016	0.58691	0.00907	0.0194	2.4480
2.5	0.6	0.62616	0.40853	0.01646	0.3210	2.5454
1.2	2.5	0.58607	0.36570	0.02222	0.2465	2.4623

Table 6: $E(XY)$, covariance, and correlation for sets A, B, and C.

Set	λ	$E(XY)$	Cov	Corr
A	-1.0	0.325735	-0.006708	-0.3230
	-0.5	0.329089	-0.003354	-0.1615
	0.5	0.335796	0.003354	0.1615
	1.0	0.339150	0.006708	0.3230
B	-1.0	0.374978	-0.005100	-0.3238
	-0.5	0.377528	-0.002550	-0.1619
	0.5	0.382628	0.002550	0.1619
	1.0	0.385178	0.005100	0.3238
C	-1.0	0.360773	-0.006198	-0.3242
	-0.5	0.363872	-0.003099	-0.1621
	0.5	0.370070	0.003099	0.1621
	1.0	0.373169	0.006198	0.3242

Table 7: $E(Y \mid X = x)$ and $\mathrm{Var}(Y \mid X = x)$ for set A. Format: Mean (Variance).

λ	$x = 0.2$		$x = 0.4$		$x = 0.6$		$x = 0.8$	
-0.8	0.6579	(0.01691)	0.6390	(0.01842)	0.5787	(0.01845)	0.5412	(0.01480)
0.0	0.5950	(0.01916)	0.5950	(0.01916)	0.5950	(0.01916)	0.5950	(0.01916)
0.8	0.5321	(0.01349)	0.5510	(0.01603)	0.6113	(0.01934)	0.6488	(0.01773)

Table 8: $E(X \mid Y = y)$ and $\mathrm{Var}(X \mid Y = y)$ for set A. Format: Mean (Variance).

λ	$y = 0.2$		$y = 0.4$		$y = 0.6$		$y = 0.8$	
-0.8	0.6269	(0.02000)	0.6181	(0.02085)	0.5524	(0.02227)	0.5026	(0.01760)
0.0	0.5587	(0.02251)	0.5587	(0.02251)	0.5587	(0.02251)	0.5587	(0.02251)
0.8	0.4905	(0.01571)	0.4994	(0.01712)	0.5650	(0.02267)	0.6148	(0.02112)

The correlation is almost exactly proportional to λ (approximately 0.324λ) in all three sets, causing the lines to overlap in the graphical illustration (see Figure 5).

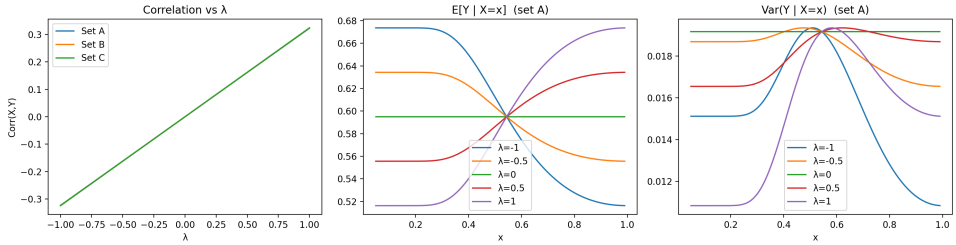


Figure 5: Correlation against λ for the three sets, the regression $E(Y \mid X = x)$, and the conditional variance $\mathrm{Var}(Y \mid X = x)$.

5 Bivariate Order Statistics of the BGUT Distribution

Let $(X_1, Y_1), \ldots, (X_n, Y_n)$ be an iid sample from BGUT. Order the pairs by X, giving $X_{1:n} \leq \cdots \leq X_{n:n}$. The Y that comes with $X_{r:n}$ is called the **concomitant** $Y_{[r:n]}$. This pair $(X_{r:n}, Y_{[r:n]})$ is the bivariate order statistic, and it is the structure that ranked set sampling (RSS) uses later. At the end of this section, we also cover the componentwise extremes.

5.1 Order Statistics of X

Since $F_1 = G_1^{\alpha_1}$, the maximum of i iid GUT(θ_1, α_1) variables is GUT$(\theta_1, i\alpha_1)$. Combining this with the standard identity for order statistic moments [**?**] gives:

$$\mu_{r:n}^{(k)} = E(X_{r:n}^k) = \sum_{i=r}^{n} (-1)^{i-r} \binom{i-1}{r-1} \binom{n}{i} \mu_k(\theta_1, i\alpha_1) \tag{20}$$

where $\mu_k(\theta, \alpha)$ is the GUT moment from Section 4. At $\alpha_1 = 1$, this reduces to Theorem 2.1 of Akhter et al. [2026].

5.2 Concomitant and Joint Distribution

Joint pdf of $(X_{r:n}, Y_{[r:n]})$:

$$f_{r:n}(x, y) = f_{r:n}(x)\, f(y \mid x) = \frac{n!}{(r-1)!(n-r)!} F_1^{r-1}(1-F_1)^{n-r} f_1(x)\, f_2(y) \big[1+\lambda(1-2F_1(x))(1-2F_2(y))\big] \tag{21}$$

Concomitant pdf. Integrate over x, using $F_1(X_{r:n}) \sim \mathrm{Beta}(r, n-r+1)$, so that $E[F_1(X_{r:n})] = r/(n+1)$:

$$f_{[r:n]}(y) = f_2(y)\Big[1 + \lambda_r\,(1 - 2F_2(y))\Big], \qquad \lambda_r = \lambda\,\frac{n+1-2r}{n+1} \tag{22}$$

Thus, the concomitant is again an FGM-type density with an effective parameter λ_r. It equals $f_2(y)$ at the middle rank (when n is odd and $r = (n+1)/2$), and it shifts toward larger y for large r if $\lambda > 0$.

Concomitant moments:

$$E\big(Y_{[r:n]}^s\big) = \mu_s^Y + \lambda_r A_s^Y, \qquad A_s^Y = \mu_s(\theta_2, \alpha_2) - \mu_s(\theta_2, 2\alpha_2) \tag{23}$$

Joint moments, using $E[X_{r:n}F_1(X_{r:n})] = \frac{r}{n+1}\mu_{r+1:n+1}$:

$$E\big(X_{r:n}^a Y_{[r:n]}^b\big) = \mu_{r:n}^{(a)}\mu_b^Y + \lambda A_b^Y\Big[\mu_{r:n}^{(a)} - \tfrac{2r}{n+1}\mu_{r+1:n+1}^{(a)}\Big] \tag{24}$$

Covariance and correlation:

$$\mathrm{Cov}(X_{r:n}, Y_{[r:n]}) = \frac{2r\lambda A_1^Y}{n+1}\Big[\mu_{r:n} - \mu_{r+1:n+1}\Big], \qquad \rho_{r:n} = \frac{\mathrm{Cov}(X_{r:n}, Y_{[r:n]})}{\sqrt{\mathrm{Var}(X_{r:n})\,\mathrm{Var}(Y_{[r:n]})}} \tag{25}$$

The bracket is negative and A_1^Y is negative, so the sign of the covariance equals the sign of λ.

5.3 Componentwise Extremes

For the vector of componentwise maxima $(X_{n:n}, Y_{n:n})$, the joint cdf is $F(x,y)^n$. For the minima, the joint survival function is $S(x,y)^n$. Joint moments follow from $E[M_X M_Y] = \iint [1 - F_1^n - F_2^n + F^n]\,dx\,dy$, evaluated by numerical integration (see Table 13).

5.4 Verification

The closed-form order statistic moments match numerical integration to 8 decimal places. For example, $\mu_1(3:5) = 0.54952906$ and $\mu_2(5:5) = 0.56307650$ in both. The identities $\sum_r \mu_{r:n} = nE(X)$ and $\sum_r E[Y_{[r:n]}] = nE(Y)$ hold exactly (2.79366428 and 2.97496821 respectively for $n = 5$). The concomitant pdf integrates to 1.

A Monte Carlo check with 300,000 samples of size $n = 5$ (Set A, $\lambda = 0.8$) agrees with the closed forms, as shown in Table 9. The largest gaps are in the $r = 4$ covariance and correlation, which are within about two Monte Carlo standard errors. For the componentwise maximum with $n = 5$, the numerical correlation is 0.1104 and the Monte Carlo value is 0.1108.

Table 9: Monte Carlo verification of concomitant moments ($n = 5$, Set A, $\lambda = 0.8$).

r	$E[Y_{[r:5]}]$ (Closed / MC)	Cov (Closed / MC)	Corr (Closed / MC)
1	0.55303 / 0.55330	0.001283 / 0.001281	0.1352 / 0.1352
2	0.57401 / 0.57415	0.001743 / 0.001745	0.1659 / 0.1663
3	0.59499 / 0.59538	0.002014 / 0.002022	0.1709 / 0.1715
4	0.61598 / 0.61627	0.002091 / 0.002131	0.1585 / 0.1612
5	0.63696 / 0.63712	0.001813 / 0.001799	0.1241 / 0.1231

5.5 Table Values

We evaluate the moments for Set A: $(\theta_1, \alpha_1, \theta_2, \alpha_2) = (1.5, 1.2, 2.0, 0.8)$. Table 10 presents the order statistics of X. Means increase with r and with n for the maximum, and variances decrease as n grows for a fixed r. Table 11 displays the concomitant means, variances, covariances, and correlations. At the middle rank of an odd n (e.g., $n = 3, r = 2$), $E[Y_{[r:n]}] = E(Y) = 0.59499$, as the theory predicts.

Table 12 shows the correlation $\mathrm{Corr}(X_{r:5}, Y_{[r:5]})$ for several values of λ. The correlation is largest in magnitude at the middle ranks and smaller at the extremes. Table 13 displays the componentwise extremes. The correlation between the extremes shrinks as n grows, which is expected for the FGM copula, whose tail dependence is zero.

Table 10: Order statistics of $X \sim \mathrm{GUT}(1.5, 1.2)$.

n	r	$\mu_{r:n}$	$\mu^2_{r:n}$	Var
1	1	0.55873	0.33469	0.02251
2	1	0.47349	0.23675	0.01256
	2	0.64398	0.43263	0.01792
3	1	0.43422	0.19724	0.00869
	2	0.55202	0.31578	0.01106
	3	0.68996	0.49106	0.01501
4	1	0.41049	0.17522	0.00672
	2	0.50541	0.26329	0.00785
	3	0.59862	0.36827	0.00992
	4	0.72041	0.53199	0.01300
5	1	0.39411	0.16086	0.00553
	2	0.47600	0.23266	0.00608
	3	0.54953	0.30923	0.00725
	4	0.63135	0.40762	0.00902
	5	0.74268	0.56308	0.01151

Table 11: Concomitants for Set A with $\lambda = 0.8$.

n	r	E	Var	Cov	Corr
2	1	0.57401	0.01815	+0.003296	+0.2183
	2	0.61598	0.01929	+0.003859	+0.2076
3	1	0.56352	0.01731	+0.002241	+0.1827
	2	0.59499	0.01916	+0.002934	+0.2016
	3	0.62647	0.01903	+0.002875	+0.1701
4	1	0.55723	0.01671	+0.001650	+0.1557
	2	0.58240	0.01866	+0.002222	+0.1836
	3	0.60758	0.01935	+0.002472	+0.1784
	4	0.63276	0.01876	+0.002242	+0.1436
5	1	0.55303	0.01626	+0.001283	+0.1352
	2	0.57401	0.01815	+0.001743	+0.1659
	3	0.59499	0.01916	+0.002014	+0.1709
	4	0.61598	0.01929	+0.002091	+0.1585
	5	0.63696	0.01854	+0.001813	+0.1241

Table 12: $\mathrm{Corr}(X_{r:5}, Y_{[r:5]})$ for several values of λ.

λ	$r = 1$	$r = 2$	$r = 3$	$r = 4$	$r = 5$
-1.0	-0.1614	-0.2017	-0.2136	-0.2065	-0.1726
-0.5	-0.0778	-0.1004	-0.1068	-0.1008	-0.0793
0.5	0.0809	0.1023	0.1068	0.0989	0.0763
1.0	0.1761	0.2096	0.2136	0.1986	0.1582

Table 13: Componentwise extremes for Set A.

λ	n	$E[\max X]$	$E[\max Y]$	Corr(max)	$E[\min X]$	$E[\min Y]$	Corr(min)
-0.8	2	0.64398	0.67368	-0.1802	0.47349	0.51631	-0.1840
	3	0.68996	0.71593	-0.1401	0.43422	0.47988	-0.1441
	5	0.74268	0.76426	-0.0981	0.39411	0.44246	-0.1005
	10	0.80394	0.82036	-0.0570	0.35228	0.40312	-0.0569
0.8	2	0.64398	0.67368	0.1976	0.47349	0.51631	0.2085
	3	0.68996	0.71593	0.1573	0.43422	0.47988	0.1678
	5	0.74268	0.76426	0.1104	0.39411	0.44246	0.1163
	10	0.80394	0.82036	0.0626	0.35228	0.40312	0.0630

6 Parameter Estimation under Simple Random Sampling (SRS)

In this section, we discuss the maximum likelihood estimation (MLE) of the BGUT distribution parameters under simple random sampling (SRS). Model adequacy is assessed using information criteria, likelihood-ratio tests, and goodness-of-fit statistics.

6.1 Data and Method

For illustration, we simulate a dataset of $n = 200$ pairs from the BGUT distribution with known true parameters $(\theta_1, \alpha_1, \theta_2, \alpha_2, \lambda) = (1.5, 1.2, 2.0, 0.8, 0.8)$ using seed 2024. The sample is saved as `bgut_sample_data.csv`. The sample summary statistics are: $\bar{x} = 0.5371$ (sd $= 0.1483$), $\bar{y} = 0.5831$ (sd $= 0.1321$), Pearson's $r = 0.1772$, and Spearman's $\rho = 0.2028$.

The log-likelihood function for the BGUT distribution is given by:

$$\ell(\boldsymbol{\Theta}) = \sum_{i=1}^{n} \Big[\ln f_1(x_i) + \ln f_2(y_i) + \ln \big(1 + \lambda(1 - 2u_i)(1 - 2v_i)\big) \Big] \tag{26}$$

where $u_i = F_1(x_i)$ and $v_i = F_2(y_i)$. The log-likelihood is maximized using the L-BFGS-B algorithm followed by Nelder–Mead, starting from values based on the fitted marginals plus 6 random starts. The information criteria are calculated as AIC $= 2k - 2\ell$, BIC $= k \ln n - 2\ell$, along with CAIC and HQIC.

6.2 Maximum Likelihood Estimates

The maximum likelihood estimates for the full BGUT model are presented in Table 14. The maximum log-likelihood achieved is 241.9985.

Table 14: Maximum likelihood estimates for the full BGUT model.

Parameter	MLE	True
θ_1	2.17713	1.5
α_1	0.38811	1.2
θ_2	2.43099	2.0
α_2	0.44295	0.8
λ	0.62797	0.8

6.3 Model Comparison

We compare the full BGUT model against two nested models: the independent GUT model ($\lambda = 0$) and the bivariate UT-FGM model ($\alpha_1 = \alpha_2 = 1$). The results are shown in Table 15. AIC picks the full BGUT model, while BIC picks the bivariate UT-FGM model due to its heavier penalty on the two extra parameters. The AIC difference between the two is only 0.55.

Table 15: Model comparison using information criteria.

Model	k	logL	AIC	BIC	CAIC	HQIC
Independent GUT ($\lambda = 0$)	4	237.684	−467.367	−454.174	−450.174	−462.028
Bivariate UT-FGM ($\alpha_1 = \alpha_2 = 1$)	3	239.721	−473.443	−463.548	−460.548	−469.438
BGUT (full)	5	241.998	−473.997	−457.505	−452.505	−467.323

6.4 Likelihood-Ratio Tests

Likelihood-ratio (LR) tests are conducted to evaluate the necessity of the dependence and generalization parameters (see Table 16). The test for independence ($\lambda = 0$) is rejected at the 1% significance level. However, the test for the generalization parameters ($\alpha_1 = \alpha_2 = 1$) yields a p-value of 0.10257, indicating that the α parameters are not significantly different from 1 at the 5% level for this dataset.

Table 16: Likelihood-ratio tests for nested hypotheses.

Hypothesis	LR	df	p-value
$\lambda = 0$ (independence)	8.6297	1	0.00331
$\alpha_1 = \alpha_2 = 1$ (UT-FGM)	4.5544	2	0.10257

6.5 Goodness-of-Fit

The goodness-of-fit is assessed using the Kolmogorov–Smirnov (KS) statistic. Table 17 presents the KS statistics and p-values for the marginals, the conditional distribution via the Rosenblatt transform, and the joint distribution. All four tests show no evidence against the full BGUT model. The bootstrap p-values, derived from 200 parametric-bootstrap refits, are the reliable ones, as the asymptotic p-values ignore parameter estimation and are conservative.

Table 17: Goodness-of-fit for the full BGUT model.

Test	KS stat.	p (asymp.)	p (boot.)
X marginal vs GUT(θ_1, α_1)	0.0444	0.8095	0.3333
Y marginal vs GUT(θ_2, α_2)	0.0322	0.9815	0.8955
Conditional $F(Y \mid X) \sim U(0,1)$	0.0286	0.9953	0.9751
Joint $\sup \lvert F_n(x,y) - F(x,y) \rvert$	0.0519	—	0.4726

Additionally, we compute the KS statistics for the marginals under the competing models (Table 18). The full BGUT model provides the best fit for both marginals.

Table 18: KS tests for the marginals under competing models.

Model	X marginal		Y marginal	
	D	p-value	D	p-value
Independent GUT ($\lambda = 0$)	0.0469	0.7529	0.0316	0.9850
Bivariate UT-FGM ($\alpha_1 = \alpha_2 = 1$)	0.0678	0.3035	0.0517	0.6392
BGUT (full)	0.0444	0.8095	0.0322	0.9815

6.6 Remarks on Parameter Identifiability

It is worth noting that the α estimates ($\hat{\alpha}_1 = 0.39$, $\hat{\alpha}_2 = 0.44$) are relatively far from the true values (1.2 and 0.8). This is not a bug, but rather a feature of weak identifiability between θ and α at $n = 200$; they trade off against each other along a flat ridge of the likelihood. For instance, fixing α_1 and refitting gives profile log-likelihoods of 241.997 ($\alpha_1 = 0.4$), 241.125 (0.8), 240.133 (1.2), and 238.68 (2.0). The log-likelihood at the true parameters is 238.23, compared to 241.998 at the MLE. This matches the non-significant LR test for $\alpha_1 = \alpha_2 = 1$. Conversely, $\hat{\lambda} = 0.628$ is near the true value of 0.8, and the data's Spearman correlation of 0.203 is close to the theoretical $\lambda/3 = 0.267$. The Monte Carlo study in Section 8 will illustrate how this behavior evolves as n grows.

7 SRS Inference: Observed Information, Confidence Intervals, and Graphical Analysis

The data used in this section are the same simulated sample as before ($n = 200$, true values $(1.5, 1.2, 2.0, 0.8, 0.8)$).

7.1 Observed Fisher Information

The observed information matrix is the Hessian of the negative log-likelihood at the MLE, $\mathbf{J}(\hat{\boldsymbol{\Theta}}) = -\partial^2\ell/\partial\boldsymbol{\Theta}\partial\boldsymbol{\Theta}^\top$. It was computed using central differences, with two different step sizes agreeing to a relative difference of 4.97×10^{-6}. The matrix is presented in Table 19.

All five eigenvalues are positive $(4.396, 6.021, 25.775, 1247.005, 1850.539)$, confirming that the MLE is a proper local maximum. The condition number is 421.

Table 19: Observed Fisher information matrix $\mathbf{J}(\hat{\boldsymbol{\Theta}})$.

	θ_1	α_1	θ_2	α_2	λ
θ_1	376.669	713.754	−41.445	−87.025	6.229
α_1	713.754	1380.516	−87.055	−182.619	11.356
θ_2	−41.445	−87.055	291.514	550.168	5.735
α_2	−87.025	−182.619	550.168	1059.052	9.652
λ	6.229	11.356	5.735	9.652	25.984

7.2 Standard Errors and Correlation of Estimators

The standard errors are obtained from the square roots of the diagonal elements of $\mathbf{J}^{-1}$: $\mathrm{SE}(\hat{\theta}_1) = 0.3634$, $\mathrm{SE}(\hat{\alpha}_1) = 0.1903$, $\mathrm{SE}(\hat{\theta}_2) = 0.4213$, $\mathrm{SE}(\hat{\alpha}_2) = 0.2215$, and $\mathrm{SE}(\hat{\lambda}) = 0.1973$.

The correlation matrix of the estimators (Table 20) reveals the main structural feature of this model: θ and α within each marginal have an estimated correlation of -0.990. This confirms the flat ridge in the likelihood surface flagged in the previous section. Conversely, the dependence parameter λ is almost uncorrelated with the marginal parameters, meaning the dependence and marginal parts of the model separate cleanly.

Table 20: Correlation matrix of the estimators.

	θ_1	α_1	θ_2	α_2	λ
θ_1	1.000	−0.990	0.044	−0.055	−0.022
α_1	−0.990	1.000	−0.056	0.070	0.013
θ_2	0.044	−0.056	1.000	−0.990	−0.054
α_2	−0.055	0.070	−0.990	1.000	0.045
λ	−0.022	0.013	−0.054	0.045	1.000

7.3 Confidence Intervals (95%)

Because of the high correlation between θ and α, the standard Wald intervals are unreliable. We compute four types of intervals: plain Wald, log/Fisher-z Wald, profile likelihood, and bootstrap percentile ($B = 300$). The results are shown in Table 21.

Table 21: Standard errors and 95% confidence intervals.

Param	MLE	SE	Wald	Log/Fisher-z	Profile LR	Boot pct ($B = 300$)
θ_1	2.1771	0.3634	[1.465, 2.889]	[1.570, 3.020]	[1.425, 2.856]	[1.547, 2.893]
α_1	0.3881	0.1903	[0.015, 0.761]	[0.148, 1.014]	[0.174, 1.225]	[0.160, 0.995]
θ_2	2.4310	0.4213	[1.605, 3.257]	[1.731, 3.414]	[1.560, 3.219]	[1.768, 3.540]
α_2	0.4430	0.2215	[0.009, 0.877]	[0.166, 1.180]	[0.189, 1.447]	[0.141, 1.083]
λ	0.6280	0.1973	[0.241, 1.015]	[0.099, 0.880]	[0.216, 0.970]	[0.216, 1.000]

The plain Wald interval yields an upper limit for λ above 1 (1.015) and a lower limit for α_1 and α_2 close to 0, both falling outside the valid parameter space. The profile likelihood intervals are asymmetric and are the most trustworthy here. These intervals contain all five true values, although $\alpha_1 = 1.2$ only just falls within the upper limit of 1.225. Both α intervals under the profile and bootstrap methods contain 1, which agrees with the non-significant likelihood-ratio test for $\alpha_1 = \alpha_2 = 1$ ($p = 0.103$). The bootstrap upper limit for λ is 1.000, indicating the estimate hits the parameter boundary in some replications.

7.4 Graphical Analysis

The behavior of the estimators and the likelihood surface are illustrated through profile likelihood and bootstrap diagnostics.

Figure 6 displays the profile deviance for each of the five parameters, along with the 95% cutoff line (3.841). The profile curves are visibly skewed for α_1 and α_2, confirming the necessity of the asymmetric profile likelihood intervals over the symmetric Wald intervals.

Figure 7 shows the bootstrap scatter plots for (α_1, θ_1) and (α_2, θ_2). These plots directly visualize the extreme ridge effect (correlation of -0.99) between the scale and shape parameters. The figure also displays the bootstrap distribution of $\hat{\lambda}$, which is slightly skewed and approaches the upper boundary.

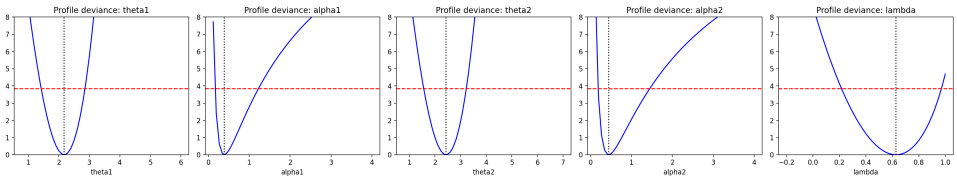


Figure 6: Profile deviance for the 5 parameters with the 95% cutoff (3.841).

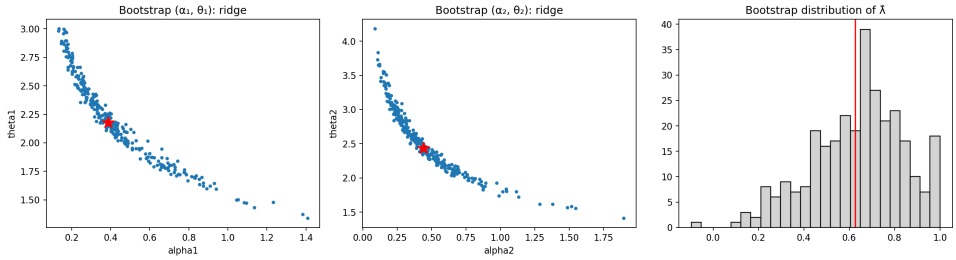


Figure 7: Bootstrap scatter plots for (α_1, θ_1) and (α_2, θ_2) showing the ridge, and the bootstrap distribution of $\hat{\lambda}$.

8 Monte Carlo Simulation Study under SRS

To evaluate the finite-sample performance of the maximum likelihood estimators under simple random sampling (SRS), a Monte Carlo simulation study was conducted.

8.1 Simulation Design

The study uses two parameter sets: Set A $= (\theta_1, \alpha_1, \theta_2, \alpha_2, \lambda) = (1.5, 1.2, 2.0, 0.8, 0.8)$ and Set B $= (1.0, 2.0, 3.0, 1.5, -0.5)$. Samples of sizes $n = 30, 50, 100$, and 200 were generated using the conditional-quantile method from Section 3. For each combination of parameter set and sample size, $R = 500$ replications were performed.

The maximum likelihood estimates were obtained using the L-BFGS-B algorithm with multiple starting points and parameter bounds $(\theta, \alpha \in [0.05, 30],\ \lambda \in [-1, 1])$. The performance is measured using the median of the estimates, bias, mean squared error (MSE), mean relative error (MRE), and the coverage probability (CP) of 95% confidence intervals. Two types of intervals are evaluated: the plain Wald interval (CP-W) and a transformed Wald interval (CP-T), which uses a log scale for θ and α, and a Fisher-z transformation for λ.

8.2 Boundary Issues and Valid Intervals

Because the likelihood surface exhibits a flat ridge between θ and α (as noted in Section 7), the optimizer occasionally lands on the edge of the parameter space at small sample sizes. An estimate is flagged as "on a bound" if any parameter hits its allowed boundary. Table 22 shows

the percentage of replications hitting a bound and the percentage of replications yielding a valid, positive-definite observed information matrix (required for Wald intervals).

Table 22: Percentage of estimates on a parameter bound and valid information matrix.

Set	n	Estimates on a bound (%)	Valid information matrix (%)
A	30	47.2	52.8
	50	42.8	57.2
	100	26.0	74.0
	200	16.4	83.6
B	30	42.2	57.8
	50	27.4	72.6
	100	9.6	90.4
	200	2.0	98.0

Set B is less prone to the bound problem than Set A (e.g., 2% vs 16.4% at $n = 200$), likely because its α values are larger and the likelihood is less flat in that region.

8.3 Simulation Results

The simulation results for Sets A and B are presented in Tables 23 and 24, respectively.

Table 23: Monte Carlo results for Set A: $(1.5, 1.2, 2.0, 0.8, 0.8)$.

n	Param	Median	Bias	MSE	MRE	CP-W	CP-T
30	θ_1	1.8482	0.3238	1.1441	0.5791	0.939	0.830
	α_1	0.7072	3.3718	90.983	3.5938	0.697	0.886
	θ_2	2.3039	0.3858	1.7608	0.5080	0.947	0.845
	α_2	0.5463	2.5384	63.886	3.9083	0.708	0.902
	λ	0.8228	−0.1198	0.1628	0.3673	0.970	0.905
50	θ_1	1.6285	0.1533	0.6117	0.4075	0.948	0.874
	α_1	0.9363	2.2634	54.213	2.4917	0.776	0.920
	θ_2	2.1843	0.2171	1.0618	0.3986	0.927	0.864
	α_2	0.6392	2.2239	57.585	3.4049	0.738	0.902
	λ	0.8319	−0.0567	0.0853	0.2881	0.976	0.913
100	θ_1	1.5947	0.0918	0.3389	0.3068	0.951	0.895
	α_1	1.0505	1.2250	22.237	1.5285	0.795	0.922
	θ_2	2.1391	0.1449	0.4579	0.2620	0.941	0.905
	α_2	0.6470	0.6257	12.229	1.2986	0.822	0.927
	λ	0.8013	−0.0302	0.0486	0.2219	0.984	0.903
200	θ_1	1.5356	0.0604	0.1495	0.2001	0.938	0.921
	α_1	1.1325	0.3043	2.4218	0.6415	0.849	0.933
	θ_2	2.0409	0.0561	0.2039	0.1792	0.931	0.935
	α_2	0.7598	0.1904	1.1986	0.6112	0.859	0.945
	λ	0.8018	−0.0154	0.0287	0.1728	0.974	0.921

Table 24: Monte Carlo results for Set B: $(1.0, 2.0, 3.0, 1.5, -0.5)$.

n	Param	Median	Bias	MSE	MRE	CP-W	CP-T
30	θ_1	1.1970	0.3079	0.9109	0.7440	0.917	0.803
	α_1	1.3047	4.9097	141.80	3.2557	0.592	0.862
	θ_2	3.3566	0.6250	5.3001	0.5998	0.927	0.824
	α_2	1.1034	3.9059	104.50	3.3342	0.699	0.869
	λ	−0.5344	0.0175	0.2042	0.7284	0.962	1.000
50	θ_1	1.1724	0.1891	0.5048	0.5709	0.909	0.835
	α_1	1.4338	3.7087	99.545	2.5685	0.683	0.860
	θ_2	3.3377	0.3823	3.1076	0.4746	0.942	0.871
	α_2	1.1525	3.0492	76.645	2.6953	0.705	0.887
	λ	−0.4929	0.0367	0.1536	0.6376	0.964	0.986
100	θ_1	1.1035	0.0919	0.2141	0.3771	0.956	0.903
	α_1	1.6127	1.9609	43.114	1.5619	0.774	0.918
	θ_2	3.1446	0.1889	1.5554	0.3285	0.958	0.905
	α_2	1.3359	1.7896	37.883	1.7080	0.785	0.923
	λ	−0.5298	−0.0097	0.0807	0.4589	0.947	0.960
200	θ_1	1.0213	0.0399	0.1006	0.2524	0.957	0.933
	α_1	1.8914	1.0316	18.159	0.9380	0.855	0.943
	θ_2	3.0463	0.0872	0.8335	0.2416	0.931	0.916
	α_2	1.4396	0.8741	14.062	0.9968	0.843	0.931
	λ	−0.4976	0.0058	0.0389	0.3139	0.951	0.961

8.4 Discussion and Caveats

The simulation results reveal several key aspects of the BGUT model's finite-sample behavior:

- **Well-behaved parameters (θ and λ):** The bias, MSE, and MRE for θ_1, θ_2, and λ consistently decrease as n grows. For example, in Set A, the MSE of $\hat{\theta}_1$ drops from 1.1441 ($n = 30$) to 0.1495 ($n = 200$), and the bias of $\hat{\lambda}$ shrinks from -0.1198 to -0.0154. The plain Wald intervals cover θ at 93–96% for $n \geq 100$, and cover λ at 95–98%.
- **Difficulty in estimating α:** The shape parameters α_1 and α_2 are notably harder to estimate. The MSE is very large (up to 141.8 in Set B) and the mean is inflated far above the truth. This occurs because the estimator sometimes runs to the upper bound along the θ–α ridge. However, the *medians* are much closer to the true values (e.g., the median of $\hat{\alpha}_1$ in Set A rises from 0.7072 to 1.1325 against a true value of 1.2). The bias and MSE still decrease with n, but α requires larger samples for precise estimation.
- **Interval coverage:** Plain Wald intervals for α undercover significantly (59–86%). The transformed intervals improve coverage to 86–95%, but they become extremely wide at small n. The transformed intervals for θ are too narrow at small n (80–87% coverage for $n \leq 50$) but improve to 92–93% at $n = 200$. The Fisher-z interval for λ covers 90–92% in Set A and 96–100% in Set B.

It should be noted that the coverage probabilities are computed only on replications with an interior MLE and a positive-definite information matrix (53–98% of replications, see Table 22). This exclusion of the worst replications slightly flatters the intervals. Because of the weak identifiability at small n, profile-likelihood or bootstrap intervals (Section 7) are recommended over Wald intervals for α when $n \leq 50$.

9 Parameter Estimation under Ranked Set Sampling (RSS)

In this section, we discuss the maximum likelihood estimation of the BGUT parameters under ranked set sampling (RSS). Model adequacy is similarly assessed using information criteria, likelihood-ratio tests, and goodness-of-fit statistics based on the probability integral transform (PIT).

9.1 Sampling Scheme and Likelihood

To compare RSS with SRS fairly, we use a simulated sample of the same size $n = 200$. The RSS design consists of a set size $m = 5$ and $k = 40$ cycles, so $n = m \times k = 200$ measured pairs. Ranking is performed on X with Y as the concomitant. In each cycle, for $r = 1, \ldots, 5$, a fresh set of 5 pairs is drawn, ranked by X, and the r-th ranked pair $(X_{r:5}, Y_{[r:5]})$ is measured. We assume perfect ranking on X. The data are simulated from BGUT with true values $(1.5, 1.2, 2.0, 0.8, 0.8)$. The sample summary statistics are: $\bar{x} = 0.5568$, $\bar{y} = 0.6037$, and Spearman's $\rho = 0.2301$.

Each measured pair has the joint pdf of $(X_{r:m}, Y_{[r:m]})$ from Section 5, and the pairs are independent. The log-likelihood function under RSS is:

$$\begin{aligned}\ell_{RSS}(\boldsymbol{\Theta}) = \sum_{i=1}^{n} \Big[\ln \tfrac{m!}{(r_i-1)!(m-r_i)!} + (r_i - 1)\ln u_i + (m - r_i)\ln(1 - u_i) \\ + \ln f_1(x_i) + \ln f_2(y_i) + \ln\big(1 + \lambda(1 - 2u_i)(1 - 2v_i)\big)\Big]\end{aligned} \tag{27}$$

where $u_i = F_1(x_i)$, $v_i = F_2(y_i)$, and r_i is the rank of the i-th measured pair. The information criteria are calculated as $\text{AIC} = 2k - 2\ell$ and $\text{BIC} = k \ln n - 2\ell$.

9.2 Maximum Likelihood Estimates

The maximum likelihood estimates for the full BGUT model under RSS are presented in Table 25. The maximum log-likelihood achieved is 315.1772.

Table 25: Maximum likelihood estimates under RSS for the full BGUT model.

Parameter	MLE	True
θ_1	1.58273	1.5
α_1	1.04639	1.2
θ_2	1.81882	2.0
α_2	1.06567	0.8
λ	0.66541	0.8

9.3 Model Comparison

We compare the full BGUT model against the independent GUT ($\lambda = 0$) and the bivariate UT-FGM ($\alpha_1 = \alpha_2 = 1$) models. The results are shown in Table 26. Both AIC and BIC pick the bivariate UT-FGM model, because the full BGUT improves the log-likelihood by only 0.008 for two extra parameters.

Table 26: Model comparison under RSS using information criteria.

Model	k	logL	AIC	BIC	CAIC	HQIC
Independent GUT ($\lambda = 0$)	4	309.611	−611.222	−598.028	−594.028	−605.883
Bivariate UT-FGM ($\alpha_1 = \alpha_2 = 1$)	3	315.169	−624.337	−614.442	−611.442	−620.333
BGUT (full)	5	315.177	−620.354	−603.863	−598.863	−613.680

9.4 Likelihood-Ratio Tests

Likelihood-ratio (LR) tests are conducted to evaluate the dependence and generalization parameters (see Table 27). Independence ($\lambda = 0$) is rejected strongly at the 1% significance level. However, the test for the generalization parameters ($\alpha_1 = \alpha_2 = 1$) yields a p-value of 0.99136, indicating that the α parameters are not needed for this sample since the estimates are very close to 1.

Table 27: Likelihood-ratio tests for nested hypotheses under RSS.

Hypothesis	LR	df	p-value
$\lambda = 0$ (independence)	11.1326	1	0.00085
$\alpha_1 = \alpha_2 = 1$ (UT-FGM)	0.0174	2	0.99136

9.5 Goodness-of-Fit

Because RSS values are not iid from F_1, a plain KS test of the marginals is not valid. Instead, we apply the PIT with the exact RSS distributions. If the model is correct, the cdf of the r-th order statistic $F_{r:m}(x)$ (testing the X part), the cdf of the r-th concomitant $F_{[r:m]}(y)$ (testing the Y part), and the conditional cdf $F(y \mid x)$ (testing the dependence) should each be uniform on $(0, 1)$.

Table 28 presents the KS statistics and p-values. The bootstrap p-values are derived from 200 parametric RSS refits and are the reliable ones. There is no evidence against the model in any of the tests. The two "pooled" rows treat the RSS values as if they were iid, which they are not, so only their bootstrap p-values are meaningful.

Table 28: Goodness-of-fit for the full BGUT model under RSS.

Test	KS stat.	p (asymp.)	p (boot.)
X-part: $F_{r:m}(x) \sim U(0,1)$	0.0408	0.8802	0.6368
Y-part: $F_{[r:m]}(y) \sim U(0,1)$	0.0483	0.7203	0.2488
Conditional $F(y \mid x) \sim U(0,1)$	0.0435	0.8273	0.3831
Pooled X vs GUT(θ_1, α_1) (informal)	0.0381	0.9223	0.6169
Pooled Y vs GUT(θ_2, α_2) (informal)	0.0411	0.8746	0.4876

Additionally, we compute the KS statistics of the transforms under the competing models (Table 29). The full BGUT model and the UT-FGM model perform nearly identically.

Table 29: KS of the transforms under competing models (asymptotic p-values).

Model	X-part D (p)	Y-part D (p)	Conditional D (p)
Independent GUT ($\lambda = 0$)	0.0409 (0.878)	0.0413 (0.870)	0.0413 (0.870)
Bivariate UT-FGM ($\alpha_1 = \alpha_2 = 1$)	0.0408 (0.879)	0.0494 (0.694)	0.0450 (0.796)
BGUT (full)	0.0408 (0.880)	0.0483 (0.720)	0.0435 (0.827)

9.6 Reference Comparison with SRS

For a visual illustration, we compare the MLEs from the earlier SRS sample ($n = 200$) with the RSS sample ($n = 200$) in Table 30. The likelihood values under RSS and SRS are not directly comparable because they are likelihoods of different data structures; AIC and BIC should only be compared within a design.

Here, the RSS estimates are closer to the true values than the SRS ones, especially for α, which suffered from the θ–α ridge under SRS. However, this is only one sample from each design and is purely illustrative. The Monte Carlo study and relative efficiency section will test this properly.

Table 30: Reference comparison of MLEs: SRS vs. RSS ($n = 200$).

Parameter	True	SRS MLE	RSS MLE
θ_1	1.500	2.1771	1.5827
α_1	1.200	0.3881	1.0464
θ_2	2.000	2.4310	1.8188
α_2	0.800	0.4430	1.0657
λ	0.800	0.6280	0.6654

10 RSS Inference: Observed Information, Confidence Intervals, and Asymptotic Properties

The data used in this section are the same simulated RSS sample as before ($m = 5, k = 40, n = 200$; true values $(1.5, 1.2, 2.0, 0.8, 0.8)$).

10.1 Observed Fisher Information

The observed information matrix under RSS, $\mathbf{J}_{RSS}(\hat{\boldsymbol{\Theta}})$, is the Hessian of the negative RSS log-likelihood at the MLE. It was computed using central differences, with two step sizes agreeing to a relative difference of 5.62×10^{-6}. The matrix is presented in Table 31.

All five eigenvalues are positive $(1.415, 2.691, 30.971, 596.227, 1853.441)$, confirming a proper local maximum. The condition number is 1309.7, which is higher than the SRS value of 421, indicating a sharper curvature in some directions but a more severe ridge in others.

Table 31: Observed Fisher information matrix under RSS, $\mathbf{J}_{RSS}(\hat{\boldsymbol{\Theta}})$.

	θ_1	α_1	θ_2	α_2	λ
θ_1	1357.274	813.418	−67.937	−47.957	4.828
α_1	813.418	491.151	−42.557	−30.030	2.504
θ_2	−67.937	−42.557	421.509	276.748	5.727
α_2	−47.957	−30.030	276.748	183.750	3.243
λ	4.828	2.504	5.727	3.243	31.061

10.2 Standard Errors and Correlation of Estimators

The standard errors are obtained from $\mathbf{J}_{RSS}^{-1}$: $\mathrm{SE}(\hat{\theta}_1) = 0.3146$, $\mathrm{SE}(\hat{\alpha}_1) = 0.5233$, $\mathrm{SE}(\hat{\theta}_2) = 0.4633$, $\mathrm{SE}(\hat{\alpha}_2) = 0.7020$, and $\mathrm{SE}(\hat{\lambda}) = 0.1802$.

The correlation matrix (Table 32) shows that the θ–α ridge from SRS persists under RSS, with correlations of -0.996 and -0.994 within the marginals. The dependence parameter λ remains largely uncorrelated with the marginal parameters.

Table 32: Correlation matrix of the estimators under RSS.

	θ_1	α_1	θ_2	α_2	λ
θ_1	1.000	−0.996	0.022	−0.026	−0.038
α_1	−0.996	1.000	−0.027	0.032	0.036
θ_2	0.022	−0.027	1.000	−0.994	−0.069
α_2	−0.026	0.032	−0.994	1.000	0.064
λ	−0.038	0.036	−0.069	0.064	1.000

10.3 Confidence Intervals (95%)

We compute four types of 95% confidence intervals: plain Wald, log/Fisher-z Wald, profile likelihood, and bootstrap percentile ($B = 300$). The results are shown in Table 33.

Table 33: Standard errors and 95% confidence intervals under RSS.

Param	MLE	SE	Wald	Log/Fisher-z	Profile LR	Boot pct ($B = 300$)
θ_1	1.5827	0.3146	[0.966, 2.199]	[1.072, 2.337]	[0.930, 2.168]	[1.051, 2.219]
α_1	1.0464	0.5233	[0.021, 2.072]	[0.393, 2.788]	[0.459, 3.716]	[0.445, 2.816]
θ_2	1.8188	0.4633	[0.911, 2.727]	[1.104, 2.997]	[0.843, 2.674]	[0.995, 2.744]
α_2	1.0657	0.7020	[−0.310, 2.442]	[0.293, 3.876]	[0.369, > 4.0]	[0.339, 4.620]
λ	0.6654	0.1802	[0.312, 1.019]	[0.167, 0.893]	[0.285, 0.973]	[0.266, 1.000]

All profile and bootstrap intervals contain the true values. The plain Wald interval for α_2 yields a negative lower limit (-0.310) and the one for λ exceeds 1 (1.019), both of which are invalid. The profile-likelihood upper limit for α_2 extended beyond the evaluation grid ($\alpha \leq 4.0$). The bootstrap estimate hit a parameter bound in 4.0% of replications. As in SRS, profile or bootstrap intervals are preferred for α.

10.4 Asymptotic Properties

Under the usual regularity conditions, the RSS-MLE is consistent and asymptotically normal:

$$\sqrt{n}(\hat{\boldsymbol{\Theta}} - \boldsymbol{\Theta}) \xrightarrow{d} N\left(\mathbf{0}, \mathbf{I}_{RSS}(\boldsymbol{\Theta})^{-1}\right), \qquad \mathbf{I}_{RSS} = \frac{1}{m}\sum_{r=1}^{m} \mathbf{I}_r(\boldsymbol{\Theta}) \tag{28}$$

where $\mathbf{I}_r$ is the expected Fisher information in one pair $(X_{r:m}, Y_{[r:m]})$. The sample is independent but not identically distributed (each rank has its own distribution), which is why $\mathbf{I}_{RSS}$ is the average over the ranks.

A normality check on the bootstrap distributions (Table 34) shows that the θ estimates are close to normal at $n = 200$. However, the α estimates are strongly skewed (Shapiro–Wilk $p \approx 0$), explaining the poor performance of Wald intervals for α. The dependence parameter λ is mildly skewed.

Table 34: Normality check of bootstrap estimates under RSS.

Parameter	Skewness	Excess Kurtosis	Shapiro–Wilk p
θ_1	$+0.160$	-0.373	0.315
α_1	$+1.918$	$+6.100$	0.0000
θ_2	$+0.146$	-0.397	0.116
α_2	$+2.149$	$+5.579$	0.0000
λ	-0.502	$+0.370$	0.0001

The expected information was computed at the true parameters via numerical quadrature. The asymptotic standard errors at $n = 200$ and the asymptotic relative efficiency (ARE $=$ $\text{Var}_{SRS}/\text{Var}_{RSS}$) are presented in Table 35.

Table 35: Asymptotic standard errors and relative efficiency (ARE).

Parameter	SE (SRS)	SE (RSS)	ARE
θ_1	0.3901	0.3102	1.582
α_1	0.7988	0.6172	1.675
θ_2	0.4391	0.4370	1.010
α_2	0.4701	0.4668	1.014
λ	0.1834	0.1832	1.002

The determinant-based overall efficiency is 1.326. Ranking on X makes RSS substantially more efficient for the X marginal parameters (θ_1, α_1) by about 58–68%. It yields almost no gain for the Y marginal parameters or for λ, because the information about Y given X is the same under both designs. If Y is the primary variable of interest, one should rank on Y instead.

10.5 Graphical Analysis

Figure 8 displays the bootstrap scatter plots for (α_1, θ_1) and (α_2, θ_2), the bootstrap distribution of $\hat{\lambda}$, and a bar chart comparing the asymptotic standard errors under SRS and RSS. The scatter plots directly visualize the extreme ridge effect (correlation near -1) between the scale and shape parameters. The bar chart clearly illustrates the efficiency gain of RSS for the X marginal parameters, matching the asymptotic relative efficiency values in Table 35.

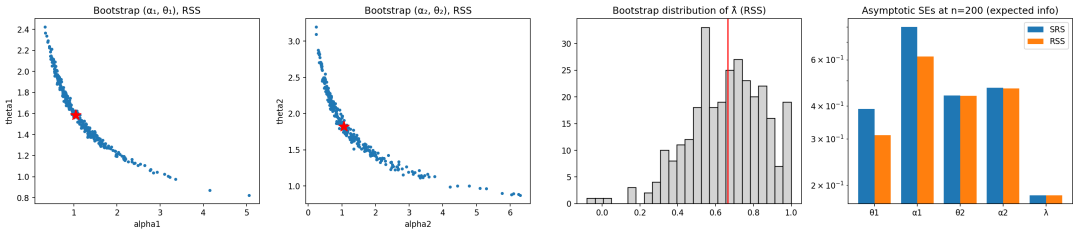


Figure 8: Bootstrap scatter plots for (α_1, θ_1) and (α_2, θ_2) showing the ridge, the bootstrap distribution of $\hat{\lambda}$, and asymptotic standard errors under SRS vs. RSS.

11 Relative Efficiency of RSS versus SRS

In this section, we compare the efficiency of ranked set sampling (RSS) to simple random sampling (SRS) for the BGUT distribution. The relative efficiency (RE) is defined as RE = Var(SRS estimator)/Var(RSS estimator), with the same number n of measured units. A value above 1 favours RSS. The design considered is ranking on X, with Y as the concomitant, for set sizes $m = 2, \ldots, 10$.

We evaluate two types of RE: (1) the RE of the sample means for $E(X)$ and $E(Y)$, which has a closed form, and (2) the asymptotic relative efficiency (ARE) of the MLEs, derived from the expected Fisher information via numerical quadrature.

11.1 RE of the Sample Means

With $n = mk$, the RSS mean has variance $\sum_r \mathrm{Var}(X_{r:m})/(m \cdot n)$ for X, and $\sum_r \mathrm{Var}(Y_{[r:m]})/(m \cdot n)$ for Y. The SRS mean has variance σ^2/n. Thus,

$$RE_X = \frac{m\,\sigma_X^2}{\sum_{r=1}^{m} \mathrm{Var}(X_{r:m})}, \qquad RE_Y = \frac{m\,\sigma_Y^2}{\sum_{r=1}^{m} \mathrm{Var}(Y_{[r:m]})} \tag{29}$$

Tables 36 and 37 present the RE for the means of X and Y across three parameter sets: A $= (1.5, 1.2, 2.0, 0.8, 0.8)$, B $= (1.0, 2.0, 3.0, 1.5, -0.5)$, and C $= (2.5, 0.6, 1.2, 2.5, 1.0)$.

Table 36: RE for the sample mean of X (the ranked variable).

m	Set A	Set B	Set C
2	1.4768	1.4737	1.4777
3	1.9429	1.9365	1.9444
4	2.4020	2.3922	2.4039
5	2.8562	2.8430	2.8583
6	3.3067	3.2901	3.3090
7	3.7544	3.7344	3.7566
8	4.1998	4.1764	4.2019
9	4.6433	4.6166	4.6452
10	5.0852	5.0552	5.0869

Table 37: RE for the sample mean of Y (the concomitant).

m	A ($\lambda = 0.8$)	B ($\lambda = -0.5$)	C ($\lambda = 1.0$)	A with $\lambda = 0$
2	1.0235	1.0091	1.0375	1.0
3	1.0357	1.0138	1.0573	1.0
4	1.0431	1.0166	1.0695	1.0
5	1.0482	1.0185	1.0779	1.0
6	1.0518	1.0198	1.0839	1.0
7	1.0545	1.0208	1.0884	1.0
8	1.0566	1.0216	1.0920	1.0
9	1.0584	1.0222	1.0949	1.0
10	1.0598	1.0227	1.0973	1.0

The ranked variable X gains substantially from RSS, with the efficiency growing roughly linearly with m (e.g., RE = 2.86 at $m = 5$ and 5.09 at $m = 10$). This gain is largely independent of the parameters. In contrast, the concomitant Y gains very little (at most 10% at $m = 10$ with $\lambda = 1$). The gain for Y depends on the dependence parameter λ; it is exactly 1.0 at independence ($\lambda = 0$) and increases with $|\lambda|$. Because the FGM copula only allows weak dependence, the benefit for Y is structurally limited. A Monte Carlo check (Set A, $m = 5, k = 5, n = 25$, 20,000 replications) confirms these closed-form values (2.9133 for X vs 2.8562; 1.0503 for Y vs 1.0482).

11.2 ARE of the Maximum Likelihood Estimators

The asymptotic relative efficiency for the MLEs is defined as $\text{ARE} = [\mathbf{I}_{SRS}^{-1}]_{jj}/[\mathbf{I}_{RSS}^{-1}]_{jj}$ for each parameter, and the overall efficiency is $(\det \mathbf{I}_{RSS}/\det \mathbf{I}_{SRS})^{1/5}$. Tables 38 and 39 present the ARE by set size m for Sets A and B.

Table 38: ARE of the MLEs by set size m for Set A.

m	θ_1	α_1	θ_2	α_2	λ	Overall
2	1.168	1.199	1.005	1.007	1.001	1.103
3	1.314	1.369	1.007	1.011	1.001	1.187
4	1.451	1.526	1.009	1.013	1.002	1.261
5	1.582	1.676	1.010	1.014	1.002	1.326
6	1.710	1.821	1.010	1.015	1.003	1.386
7	1.837	1.962	1.011	1.016	1.003	1.442
8	1.962	2.102	1.011	1.017	1.003	1.494
9	2.086	2.241	1.012	1.017	1.003	1.543
10	2.209	2.378	1.012	1.018	1.004	1.589

Table 39: ARE of the MLEs by set size m for Set B.

m	θ_1	α_1	θ_2	α_2	λ	Overall
2	1.168	1.194	1.002	1.003	1.000	1.104
3	1.314	1.361	1.003	1.004	1.000	1.189
4	1.450	1.514	1.004	1.005	1.001	1.262
5	1.581	1.660	1.004	1.006	1.001	1.328
6	1.709	1.801	1.004	1.006	1.001	1.388
7	1.834	1.939	1.005	1.006	1.001	1.443
8	1.958	2.075	1.005	1.007	1.001	1.495
9	2.081	2.210	1.005	1.007	1.001	1.544
10	2.203	2.344	1.005	1.007	1.001	1.591

Table 40 shows that the ARE barely changes across $\lambda \in [-1, 1]$. Table 41 demonstrates that ranking on Y instead of X simply reverses the pattern: the large gains shift to θ_2 and α_2, while the overall efficiency remains virtually identical.

Table 40: ARE against λ (Set A margins, $m = 5$).

λ	θ_1	α_1	θ_2	α_2	λ	Overall
-1.0	1.575	1.665	1.014	1.020	1.004	1.323
-0.5	1.589	1.684	1.004	1.006	1.001	1.329
0.0	1.592	1.688	1.000	1.000	1.000	1.330
0.5	1.589	1.684	1.004	1.006	1.001	1.329
0.8	1.582	1.676	1.010	1.014	1.002	1.326
1.0	1.575	1.665	1.014	1.020	1.004	1.323

Table 41: Ranking on X versus ranking on Y ($m = 5$).

Set	Rank on	θ_1	α_1	θ_2	α_2	λ	Overall
A	X	1.582	1.676	1.010	1.014	1.002	1.326
	Y	1.010	1.014	1.589	1.697	1.002	1.327
B	X	1.581	1.660	1.004	1.006	1.001	1.328
	Y	1.004	1.005	1.585	1.673	1.001	1.328

11.3 Discussion

The results yield several key insights:

- RSS is highly efficient for the ranked variable's parameters. The ARE for θ_1 and α_1 is 1.58 and 1.68 at $m = 5$, rising steadily to 2.21 and 2.38 at $m = 10$.
- RSS provides almost no gain for the concomitant's parameters or λ. The ARE for θ_2, α_2, and λ remains between 1.000 and 1.020 regardless of m or λ.
- The overall efficiency is 1.33 at $m = 5$ and 1.59 at $m = 10$. This averages the substantial gains for the ranked variable with the near-zero gains for the other three parameters.

- The choice of ranking variable dictates where the efficiency gains occur. If Y is the primary variable of interest, one should rank on Y to transfer the gains to θ_2 and α_2.
- The RE for the sample mean of X (2.86 at $m = 5$) is higher than the ARE for the MLEs of X (1.6 to 1.7-fold). This is because the MLE must jointly estimate the correlated θ–α parameters, whereas the sample mean utilizes the ranking information more directly.

It should be noted that these are asymptotic ($n \to \infty$) and exact-moment results. At small n, finite-sample issues like the θ–α ridge may cause the simulated MSE ratios to deviate from these ARE values, a phenomenon investigated in the following section.

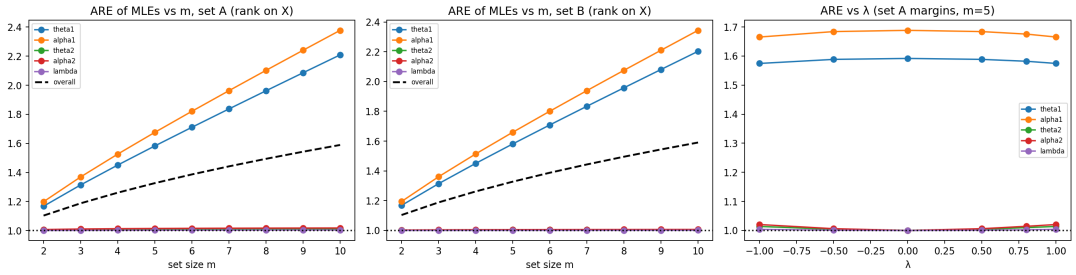


Figure 9: Asymptotic relative efficiency (ARE) of the MLEs and the overall ARE against set size m for Sets A and B, and the ARE against λ.

12 Monte Carlo Study under RSS, Comparison with SRS, and RE

To compare the finite-sample performance of the maximum likelihood estimators under ranked set sampling (RSS) and simple random sampling (SRS), a Monte Carlo simulation study was conducted.

12.1 Simulation Design

The study evaluates both RSS (set size $m = 5$, ranking on X, with Y as the concomitant) and SRS. The sample sizes are $n = 5, 10, 25, 50$, corresponding to $k = 1, 2, 5, 10$ cycles for RSS (and the same n for SRS). The simulation uses $R = 300$ replications per cell for two parameter sets: Set A $= (1.5, 1.2, 2.0, 0.8, 0.8)$ and Set B $= (1.0, 2.0, 3.0, 1.5, -0.5)$.

Efficiency is measured using the relative efficiency based on mean squared error: $\mathrm{RE}_{\mathrm{MSE}} = \mathrm{MSE(SRS)}/\mathrm{MSE(RSS)}$. A bootstrap 95% interval for each RE (1,000 resamples of the replications) is also computed to account for the noise inherent to $R = 300$.

12.2 Important Caveat about Small n

With five parameters and n of only 5 to 50 pairs, the MLE frequently lands on a bound of the parameter space (θ or α at 0.05 or 30, or λ at ± 1). This is the θ–α ridge problem from previous sections, exacerbated at small sample sizes. Table 42 shows the severity of this issue.

At $n = 5$ and $n = 10$, almost all estimates are on a bound. Consequently, those cells say little about the model's typical behavior. We recommend treating $n = 25$ and $n = 50$ as the informative cells and reporting $n = 5$ and 10 only as a stress test.

Table 42: Percentage of estimates on a parameter bound and valid information matrix.

Set	n	On a bound, SRS (%)	On a bound, RSS (%)	Valid info, SRS (%)	Valid info, RSS (%)
A	5	94.0	91.0	6.0	8.7
	10	80.0	77.0	20.0	23.0
	25	53.7	56.7	46.3	43.3
	50	43.3	36.0	56.7	64.0
B	5	92.3	89.0	7.7	11.0
	10	78.3	75.0	21.7	25.0
	25	45.7	45.3	54.3	54.7
	50	24.0	25.3	76.0	74.8

12.3 Mean Squared Errors

The MSEs for Sets A and B are presented in Tables 43 and 44. The very large α MSEs (up to about 140) stem from estimates hitting the upper bound of 30, which are far from the truth.

Table 43: MSE (SRS / RSS) for Set A.

n	θ_1	α_1	θ_2	α_2	λ
5	5.5439 / 3.6974	121.74 / 126.46	6.3922 / 6.7749	77.82 / 108.05	0.8870 / 0.7721
10	3.1222 / 1.9715	131.05 / 112.59	4.0158 / 4.0443	95.68 / 93.80	0.5447 / 0.4627
25	1.2694 / 0.8795	95.05 / 82.05	1.6219 / 1.7362	83.84 / 42.82	0.1996 / 0.1657
50	0.6923 / 0.4051	77.40 / 12.44	0.8725 / 0.8488	28.56 / 26.79	0.0917 / 0.1030

Table 44: MSE (SRS / RSS) for Set B.

n	θ_1	α_1	θ_2	α_2	λ
5	4.5654 / 2.4791	125.81 / 138.67	27.146 / 29.564	124.33 / 126.57	0.7908 / 0.7197
10	2.5688 / 1.7675	131.38 / 112.04	16.122 / 15.280	132.97 / 124.21	0.5753 / 0.5193
25	0.9628 / 0.6566	139.97 / 117.72	6.8802 / 6.7737	96.63 / 95.82	0.3095 / 0.2822
50	0.4973 / 0.2678	91.58 / 74.75	3.0972 / 2.7218	70.68 / 80.74	0.1467 / 0.1602

12.4 Relative Efficiency ($\mathrm{RE}_{\mathrm{MSE}}$)

Tables 45 and 46 present the $\mathrm{RE}_{\mathrm{MSE}}$ with 95% bootstrap intervals. Table 47 pools the RE over the four sample sizes using the geometric mean. For reference, the asymptotic values from Section 11 ($m = 5$) are $\theta_1 = 1.58$, $\alpha_1 = 1.68$ (Set A) and 1.66 (Set B), and approximately 1.00 for θ_2, α_2, and λ.

Table 45: Relative Efficiency ($\mathrm{RE}_{\mathrm{MSE}}$) with 95% bootstrap interval for Set A.

n	θ_1	α_1	θ_2	α_2	λ
5	1.499 [1.255, 1.803]	0.963 [0.671, 1.360]	0.944 [0.793, 1.139]	0.720 [0.459, 1.077]	1.149 [0.895, 1.465]
10	1.584 [1.283, 1.933]	1.164 [0.806, 1.687]	0.993 [0.850, 1.180]	1.020 [0.666, 1.570]	1.177 [0.873, 1.579]
25	1.443 [1.137, 1.852]	1.159 [0.737, 1.760]	0.934 [0.766, 1.153]	1.958 [1.170, 3.692]	1.204 [0.868, 1.623]
50	1.709 [1.358, 2.134]	6.223 [3.398, 14.068]	1.028 [0.821, 1.293]	1.066 [0.481, 2.441]	0.890 [0.644, 1.184]

Table 46: Relative Efficiency (RE_{MSE}) with 95% bootstrap interval for Set B.

n	θ_1	α_1	θ_2	α_2	λ
5	1.842 [1.479, 2.292]	0.907 [0.649, 1.270]	0.918 [0.715, 1.165]	0.982 [0.669, 1.416]	1.099 [0.918, 1.318]
10	1.453 [1.159, 1.781]	1.173 [0.800, 1.663]	1.055 [0.854, 1.279]	1.070 [0.745, 1.527]	1.108 [0.894, 1.343]
25	1.466 [1.125, 1.908]	1.189 [0.840, 1.699]	1.016 [0.797, 1.274]	1.008 [0.660, 1.585]	1.097 [0.849, 1.382]
50	1.857 [1.423, 2.426]	1.225 [0.818, 1.849]	1.138 [0.905, 1.415]	0.875 [0.537, 1.393]	0.915 [0.721, 1.127]

Table 47: Pooled relative efficiency (geometric mean of RE_{MSE}).

Set	θ_1	α_1	θ_2	α_2	λ
A	1.556	1.686	0.974	1.113	1.097
B	1.643	1.116	1.029	0.982	1.051

12.5 Discussion and Coverage

The simulation yields the following key findings:

- **The ranked variable's parameter θ_1 is estimated more efficiently under RSS.** The RE_{MSE} is between 1.44 and 1.86 in all eight cells, and every bootstrap interval excludes 1 (the lowest lower limit is 1.125). The pooled values (1.56 for A, 1.64 for B) agree closely with the asymptotic ARE of 1.58. This is the clearest result of the study.
- **θ_2 and λ show no systematic difference between the designs.** The RE for θ_2 stays between 0.93 and 1.14, and every bootstrap interval includes 1. The RE for λ ranges from 0.89 to 1.20, with all intervals including 1. This matches the theory that ranking on X adds essentially nothing for the other variable's parameters or for λ.
- **The α estimates cannot be compared reliably.** Their RE values are erratic (from 0.72 to 6.22) and their intervals are extremely wide. The value of 6.22 for α_1 in Set A at $n = 50$ stems from a few SRS estimates hitting the upper bound of 30, not from a real 6-fold advantage. The design comparison for α is thus inconclusive at these sample sizes.

Coverage of the transformed-Wald 95% intervals was also computed on replications with a valid information matrix. At $n = 50$, coverage is 84–93% for both designs (e.g., Set A, RSS: θ_1 0.844, α_1 0.870, θ_2 0.885, α_2 0.927, λ 0.859). It is similar under RSS and SRS. Values at $n = 5$ and 10 are not meaningful due to the tiny fraction of valid replications.

In summary, ranked set sampling with ranking on X improves the estimation of the ranked variable's parameter θ_1 (RE $\approx$ 1.5–1.9, consistent with the asymptotic efficiency of 1.58) and gives no meaningful gain for θ_2 and λ. The shape parameters α_1 and α_2 are poorly identified at these sample sizes under both designs, making a design comparison for them inconclusive.

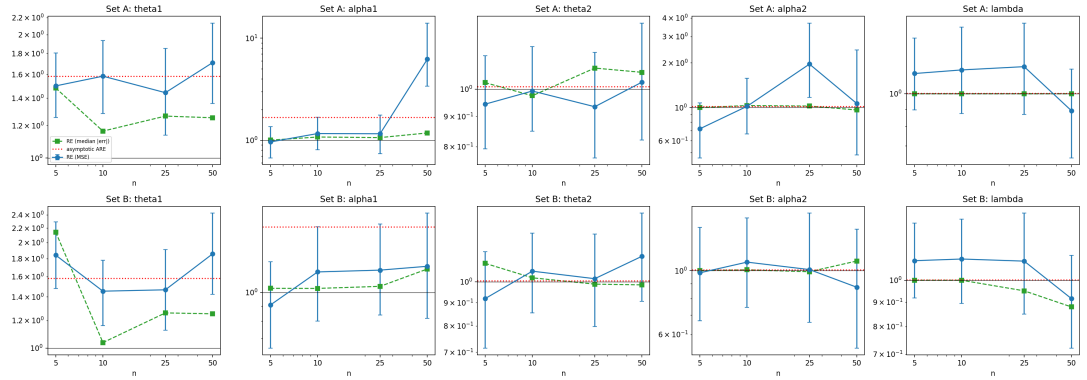


Figure 10: Relative efficiency ($\mathrm{RE}_{\mathrm{MSE}}$) with bootstrap intervals, robust RE, and asymptotic ARE against n for each parameter and both sets.

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